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Publications

Peer Reviewed

Grossi, M.D., S. Jegelka, P.F.J. Lermusiaux, T.M. Özgökmen (2025) Surface drifter trajectory prediction in the Gulf of Mexico using neural networks, *Ocean Modelling*, 196, 102543.

Shah, C, M.M. Nabi, S.Y. Alaba, M.D. Campbell, R. Caillouet, **M.D. Grossi**, J.E. Ball, and R. Moorhead (2025) YOLOv8-TF: Transformer-Enhanced YOLOv8 for Underwater Fish Species Recognition with Class Imbalance Handling, *Sensors*, 25, 1846, doi:10.3390/s25061846.

Grossi, M.D. , T.M. Özgökmen, M. Kubat (2020) Predicting particle trajectories in oceanic flows using artificial neural networks, *Ocean Modelling*, 156, 101707.

Geiger, E.F., **M.D. Grossi**, A.C. Trembanis, J.T. Kohut, M.J. Oliver (2011) Satellite-Derived Coastal Ocean and Estuarine Salinity in the Mid-Atlantic, *Continental Shelf Research*, doi:10.1016/j.csr.2011.12.001.

Conference Proceedings

Shah, C., M.M. Nabi, I.A. Ebu, J. Prior, **M.D. Grossi**, F. Wallace, T. Rowell, J.E. Ball, R. Moorhead, R. Caillouet, M. Campbell (2025) Improved fish tracking in underwater images for marine biodiversity monitoring, Proc. SPIE 13460, Machine Learning from Challenging Data 2025, 134600F (29 May 2025), https://doi.org/10.1117/12.3053499.

Shah, C., M.M. Nabi, S.Y. Alaba, R. Caillouet, J. Prior, M. Campbell, **M.D. Grossi**, F. Wallace, J.E. Ball, and R. Moorhead (2024) Active detection for fish species recognition in underwater environments, Proc. SPIE 13061, Ocean Sensing and Monitoring XVI, 130610D, 6 June 2024, https://doi.org/10.1117/12.3013344.

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Technical Reports (not peer reviewed)

Grossi, M.D., M. Monim, A. Gangopadhyay (2017) Global Climate Patterns: An Overview of Arctic Oscillation, Pacific Decadal Oscillation, Pacific/North American Pattern, and El Niño Southern Oscillation, University of Massachusetts Dartmouth School for Marine Science and Technology Technical Report SMAST-17-0401, doi:10.13140/RG.2.2.34586.44480.

W.S. Brown and **M. Grossi** (2016) Pre- and Post-Mission-6 Glider CTD Comparison Measurements: 11 June and 22 July 2015, University of Massachusetts Dartmouth School for Marine Science and Technology Technical Report SMAST-16-0501.

W.S. Brown and **M. Grossi** (2015) Pre- and Post-Mission Glider CTD Comparison Measurements: 19 June 2014 and 6 February 2015, University of Massachusetts Dartmouth School for Marine Science and Technology Technical Report SMAST-15-06-01.

Presentations | Posters | Contributed Abstracts

Grossi, M.D., M. Kubat, T.M. Özgökmen (2020) Can Neural Networks Learn Realistic Ocean Trajectories?, Gulf of Mexico Oil Spill and Ecosystem Science Conference, Tampa, FL.

Grossi, M.D., M. Kubat, T.M. Özgökmen (2020) Predicting particle trajectories using artificial neural networks, Multidisciplinary University Research Initiative (MURI) Machine Learning for Submesoscale Characterization, Ocean Prediction, and Exploration (SCOPE) project kickoff meeting, Massachusetts Institute of Technology, Cambridge, MA.

Grossi, M.D., M. Kubat, T.M. Özgökmen (2019) Predicting particle trajectories using artificial neural networks, Consortium for Advanced Research on Transport of Hydrocarbon in the Environment Fall All-Hands Meeting, University of Miami, Miami, FL.

Grossi, M.D., M. Kubat, T.M. Özgökmen (2019) Predicting Oil Transport in Oceanic Flows: Are Neural Networks Up to the Task?, Gulf of Mexico Oil Spill and Ecosystem Science Conference, New Orleans, LA.

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Treible, L.M., E.F. Geiger, **M.D. Grossi**, M.J. Oliver (2010) The Impact of Wind on Non-Photochemical Quenching in the Mid-Atlantic Coastal Ocean, Ocean Sciences Meeting, Portland, OR.

Geiger, E., **M. Grossi**, M. Oliver (2009) Developing Satellite-Derived Salinity Product for the Mid-Atlantic Coastal Region, Delaware EPSCoR Annual State Meeting, Newark, DE.

Oliver, M., **M. Grossi**, E. Geiger (2009) 3-D Mapping of Ocean Water Masses and Provinces, Joint Carbon/Biodiversity Meeting, New York, NY.

Splitt, M.E., **M.D. Grossi** (2008) Evaluation of the Real-Time Ocean Forecast System in Florida Atlantic Coastal Waters, Ocean Sciences Meeting, Orlando, FL.

Grossi, M.D. (2007) Evaluation of the Real-Time Ocean Forecast System in Florida Atlantic Coastal Waters, Florida Institute of Technology Department of Marine & Environmental Systems Summer Symposium, Melbourne, FL. (Both oral and poster presentation)